HSUAN YA (MICHAEL) TSAI

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PROFESSIONAL SUMMARY

Senior infrastructure and networking engineer with 14+ years of experience in cloud platforms and network security. Owns software-FPGA interfaces for Azure host networking and SDN, delivering features from architecture through production rollout. Expertise spans NIC integration and qualification, GPU networking enablement, and automation for production monitoring, mitigation, and testing.

TECHNICAL EXPERTISE

Languages & Scripting: C, C++, Python, PowerShell, Bash/Shell

Cloud Networking & Systems: SDN, DPDK, Azure Accelerated Networking, SR-IOV, FPGA Offload, MANA, NVIDIA/Mellanox NICs, Azure Boost, VNet Encryption, TCP/IP, Linux

Infrastructure & Tooling: CI/CD, Build & Test Automation, Deployment Automation, Production Monitoring & Mitigation, Kusto, Google Test, Jenkins

PROFESSIONAL EXPERIENCE

MICROSOFT | Azure Core Compute / Host Networking — Redmond, WA

Senior Software Engineer — Sep. 2021 - Present

Software Engineer II — Apr. 2019 - Aug. 2021

- Owned software-FPGA interfaces for Azure host networking datapath and SDN across NIC generations, from 30 Gbps Mellanox CX3Pro to 400 Gbps MANA, integrating software built on DPDK with FPGA flow offload.
- Led architecture and software/hardware interface design for per-host virtual port (vport) expansion from 128 to 512 on Microsoft Azure Network Adapter (MANA), using extensible NIC fields; delivered implementation, validation, and production rollout.
- Enabled Accelerated Networking for two newly introduced A100/H100-based GPU hardware SKUs that previously lacked support; established zero-touch onboarding for future clusters using similar SKUs.
- Owned Azure-side integration and qualification of NVIDIA/Mellanox NICs deployed across millions of Azure servers, including driver validation, rollout, telemetry analysis, and production issue mitigation.
- Automated Kusto monitoring of production crashes and unexpected component behavior, replacing daily manual queries with configurable schedules; introduced on-demand and scheduled automated mitigation based on monitoring results.
- Initiated an AI-assisted testing effort that increased line coverage for the FPGA-based offload project from 39% to 82%; updated repository AI guidelines to require automatic unit-test generation for production code changes.
- Enabled Accelerated Networking and VNet Encryption across new Azure hardware, VM, and OS platforms, including new VM generations and Azure Boost SKUs.
- Built C++ host-networking servicing components and Python/PowerShell tooling for hardware qualification and automated validation.
- Optimized build/test pipelines, reducing runtime by 60% for comparable workloads.
- Diagnosed and mitigated production networking incidents during on-call rotations, deployed fixes, and mentored junior engineers.

EARLIER EXPERIENCE

MICROSOFT | Windows Protocols — Shanghai, China
Software Engineer II — Nov. 2017 - Apr. 2019

- Developed protocol test suites in C# to validate interoperability and networking behavior across Windows releases, with continuous integration using Jenkins.

DELL INC. — Taipei, Taiwan
Senior Software Engineer — Aug. 2017 - Oct. 2017

- Developed software components for Alienware gaming PCs using C/C++ and C#.

TREND MICRO INC. — Taipei, Taiwan
Project Lead / Senior Software Engineer — Jan. 2015 - Aug. 2017
Software Engineer — Apr. 2012 - Jan. 2015

- Led a 5-person network engine team developing network security and advanced-threat detection capabilities in C/Linux.
- Built Deep Packet Inspection functionality across L2-L4 and application protocols, including TCP, UDP, ICMP, IP, and HTTP.
- Developed real-time threat-detection DSL capabilities and improved engineering infrastructure, test automation, and unit-test tooling using C/C++, Python, Perl, lex/yacc, and Google Test.

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA — Los Angeles, CA
Master of Science in Computer Science (General) — Sep. 2010 - Dec. 2011

NATIONAL CHIAO TUNG UNIVERSITY — Hsinchu, Taiwan
Master of Science in Computer Science — Sep. 2007 - Jul. 2009

NATIONAL CHIAO TUNG UNIVERSITY — Hsinchu, Taiwan
Bachelor of Science in Computer Science — Sep. 2002 - Jun. 2007